

RETAIL API INTEGRATION

- 1) Authentication
- 2) Products API
- 3) Stock API
- 4) Sales API

1) Authentication

Whenever a request is made, we use Basic Authentication, we will be sharing URL, Authentication details for the API

2) Products API

Products API is targeted to read products from POS system, first thing which needs to be done is make a request to the API

Request

```
GET
Authentication: Basic
Host: {{host}}/API/values
Content-Type: application/json
```

Request URL example : <http://{{host}}/api/values?modifiedDate=2018-01-01>

Modifieddatetime parameter is the date string to get products modified from passed value.

Below is sample request in PHP

```
<?php
$login = 'Basic Authentication user name';
$password = ' Basic Authentication Password';
$url = 'http://{host}/api/values?modifiedDate=2018-01-01';
$ch = curl_init();
curl_setopt($ch, CURLOPT_URL,$url);
curl_setopt($ch, CURLOPT_RETURNTRANSFER,1);
curl_setopt($ch, CURLOPT_HTTPAUTH, CURLAUTH_BASIC);
curl_setopt($ch, CURLOPT_USERPWD, "$login:$password");
$result = curl_exec($ch);
curl_close($ch);
echo($result);
```

Below is .net Sample Request

Prerequisite you need to add Newtonsoft.Json to your project.

```
HttpWebRequest request = (HttpWebRequest)WebRequest.Create(url);
string user = "Basic Authentication user name";
string pwd = "Basic Authentication Password";

request.Method = "GET";
request.ContentType = "application/json";
request.Timeout = 100000000;

string encode =
System.Convert.ToBase64String(System.Text.Encoding.GetEncoding("ISO-8859-
1").GetBytes(user + ":" + pwd));

request.Headers.Add("Authorization", "Basic " + encode);
HttpWebResponse response = (HttpWebResponse)request.GetResponse();
var responseText = new StreamReader(response.GetResponseStream()).ReadToEnd();
```

Response

Once request is successfully, you will get the modified products List as response in JSON format

```
[
  {
    "Category_Name1": "WINES SPIRITS BEERS",
    "Category_Name2": "ENO",
    "Category_Name3": "",
    "Category_Name4": "",
    "UnitOfProduct": "75CL",
    "SupplierProductCode": "3159560700077",
    "SupplierProductDescription": "",
    "SupplierName": "ENO",
    "Central_Product_Code": "ENO211488",
    "Product_Name": "Chablis Calv",
    "Product_Description": "Calvet Chablis 75CL",
    "Cost_Price": 10.5000,
    "SellingPrice": "16.6900",
    "Stock_Max_Level": 0,
    "ReOrder_Level": 0,
    "Catalogue_PageNumber": "",
    "Catalogue_Description": "",
    "UnitOfSale": 24864,
    "Physical_Qty": 100.0,
    "Weight": 0.0,
    "BackOrder_Qty": 0.0,
    "Is_Deleted": false,
    "CreatedBy": "0001",
    "CreatedDateTime": "2020-04-15T11:05:58",
    "ModifiedBy": "0001",
    "ModifiedDateTime": "2020-04-15T11:05:58",
    "Vat_Percentage": 20.0,
    "HasExpiry": false,
    "Br_Code": "EM",
    "subcategoryname": "WINE"
  }
]
```

Response fields

Category_Name1 : string

Category Level 1 to which product belongs to

Category_Name2 : string

Brand of the product

Subcategoryname : string

Category Level 2 to which product belongs to

Category_Name3 : string

Category Level 3 to which product belongs to

Category_Name4 : string

Category Level 4 to which product belongs to

UnitOfProduct : string

Unit of product is the type of unit product is sold in e.g Each,Pack of 6 etc,100cl

SupplierProductCode : string

Barcode of the product

SupplierProductDescription: string

Supplier description for the product

SupplierName: string

Name of the supplier

Central_Product_Code: string

SKU or unique product code, this is used for stocks and Sales

Product_Name: string

Name of the product

Product_Description: string

Description of the product.

Cost_Price: decimal

Cost price of the product.

SellingPrice : decimal

Selling price of the product.

Stock_Max_Level: int

Selling price of the product.

Catalogue_PageNumber: string

Catalogue page number of the product.

Catalogue_Description: string

Catalogue Description of the product.

Physical_Qty: decimal

Physical qty / Current stock of the product.

Weight: decimal

Weight of the product.

Backorder_qty: decimal

Back Order qty /Quantity expecting from suppliers of the product.

Is_Deleted: boolean

Confirms whether product is deleted true for Deleted, false for Active.

CreatedBy: string

User who created it.

Createddatetime: datetime

Product created date time

ModifiedBy: string

User who modified it.

Createddatetime: datetime

Product Modified date time

VAT_Percentage: decimal

VAT Percentage of the product

HasExpiry: boolean

Confirms whether Product has Expiry Date

Br_code: string

Confirms whether Product has Expiry Date

3) Stocks API

Stocks API is targeted to read Stock lines from POS system where stock is greater than 0, It gives all products which are greater than 0

Request

```
GET
Authentication: Basic
Host: {{host}}/API/Stock
Content-Type: application/json
```

Request URL : <http://{{host}}/api/Stock>

Once request is successful, you will be getting list of products in below JSON format

```
[
  {
    "ProductCode": "EN0071326",
    "StockCount": "100"
  }
]
```

Productcode: string

Product code/SKU for the product code, this will be same as Central_Product_Code on Products API

StockCount: integer

Stock count of the product

4) **Sales API**

Sales/orders API is used to post sales into POS

Request

```
POST
Authentication: Basic
Host: {{host}}/API/sales
Content-Type: application/json
```

Request URL example : <http://{{host}}/api/Sales>


```

[
{
  "Sales":
  {
    "Order_Id":1,
    "Order_date":"2018/05/11 11:37:00",
    "Customer_name":"test",
    "Customer_Lastname":"l name",
    "HouseNo":"",
    "Road":"",
    "City":"",
    "County":"",
    "Postcode":"",
    "CountryName":"",
    "Delivery_HouseNo":"",
    "Delivery_Road":"",
    "Delivery_City":"",
    "Delivery_County":"",
    "Delivery_Postcode":"",
    "Delivery_Country":"",
    "Phone":"",
    "Email":"test@gmail.com",
    "GrandTotal":"10.98",
    "Delivery_Cost":"",
    "Vat_Cost":"",
    "NumberOfItems":"3",
    "WebOrder_Ref":"Web-0001"
  },
  "Sales_Details":
  [
    {
      "ItemID":"item 1",
      "Quantity":2,
      "Product_Name":"test product",
      "UnitPrice":"4.99",
      "TotalPrice":"8.98",
      "VatPercentage":"20",
      "DiscountValue":"1.00"
    },
    {
      "ItemID":"item 2",
      "Quantity":1,
      "Product_Name":"product 2",
      "UnitPrice":"1.00",
      "TotalPrice":"1.00",
      "VatPercentage":"20",
      "DiscountValue":""
    }
  ],
  "Payments":
  {
    "Paidby":"Card",
    "Amount":"10.98"
  }
}
]

```

Request fields

a) Sales – This is Header for the Order

Order_ID: integer

This is unique reference for the Order

Order_date: string

Date and time of the Order

Customer_name: string

Name of the customer

Customer_Lastname: string

Lastname of the customer

House_No: string

House Number of the Invoice Address

Road: string

Street or Road Name of the Invoice Address

City: string

City or Town of the Invoice Address

County: string

County of the Invoice Address

Postcode: string

Postcode of the Invoice Address

CountryName: string

Countryname of the Invoice Address

Delivery_HouseNo: string

House Number of the Delivery Address

Delivery_Road: string

Street or Road Name of the Delivery Address

Delivery_City: string

City or Town of the Delivery Address

Delivery_County: string

County of the Delivery Address

Delivery_Postcode: string

Postcode of the Delivery Address

Delivery_Country: string

Country of the Delivery Address

Phone: string

Phone number of the customer

Email: string

Email address of the customer

GrandTotal: Decimal

Total of the order including delivery charges

Delivery_Cost: Decimal

Delivery charges for the order

Note: If you are passing Delivery charges as Sales Line, please pass this value as "0"

VAT_Cost: decimal

Total VAT for the order

NumberofItems: integer

Number of Items on the Order, sum of individual quantities.

WebOrder_Ref: string

In case if you have different reference for web order which is string, please pass it here for order look up, if you do not have any, please pass the OrderID value which is the first field.

b) Sales_Details – This are Individual Lines in Order

ItemID: string

ItemID or SKU is Central_product_code from Products API

Quantity: integer

Quantity of the item on the order line

ProductName: string

Product Name of the Item.

UnitPrice: decimal

Unit price of the product.

TotalPrice: decimal

Total price of the product after any discounts are applied.

VatPercentage: decimal

VAT Percentage of the product line.

Discountvalue: decimal

Discount applied on the product, this should be value discount only.

c) Payments

PaidBy: string

How the order is paid, Values can be Cash/Card/Cheque/Bank Transfer.

Amount: decimal

Amount paid for specified payment type.

Response

If the Order Request is right, you will be expecting below response

```
[  
{  
  "Order_Id":1,  
  "Status":"Success",  
  "Reason":"","  
  "WebOrderRef":"Web-0001"  
}  
]
```

Order_ID: string

It's the Order_ID passed in request

Status: string

It's the Status of the request, it will be Success/Failed.

Reason: string

Reason for failure will be provided

WebOrderRef: string

Weborder_Ref passed in request